

End-to-End and the Boardroom's Five Words

Chapter 7 (objective 1.2) and Chapter 8 (objective 1.6). Input, processing, output, the loop nobody draws, and value as a pillar with a number attached.

Objective 1.2: the restaurant, not the recipe

The picture. A recipe is processing. A restaurant is end-to-end: the door (who gets in, in what state), the kitchen, the pass where the head chef tastes before plates leave, and the part amateurs forget, the comment cards that change next month's menu. An architecture diagram without the comment cards is a recipe pretending to be a restaurant.

Stage	Kaveri concretely	The decision that lives here
Input	Email, portal and phone-transcript exceptions normalised into one ticket schema	Validation and PII scrubbing happen here, before any model, not mid-pipeline
Processing	Router (E3) to specialist chains (E2), agent tier (E7) for the tangled cases	Patterns compose; each tier is as simple as its slice allows
Output	Draft replies, re-route orders, refund recommendations	The human gate: refunds at or above Rs 10,000 pause for Meera's desk. Domain 5 will formalise it
Feedback	Dispatcher edits diffed against drafts; outcomes logged; the eval set grows weekly (Domain 4)	Without this loop the system is frozen at launch quality forever

The Atlas A3 diagram draws the four stages as bands with a purple arrow running from feedback back up to processing. The brief calls that arrow the answer to a whole family of stems: when a 1.2 option list contains add a feedback loop from dispatcher edits, check whether the described architecture already has one. If not, that option beats every clever processing upgrade on the page.

The tell

1.2 stems love an architecture that is complete except for one absent arc, usually feedback or the human gate. The right option is rarely more processing. It is the **missing arc**. Count the four stages against every option before judging any of them clever.

Everyday example: a new cafe opens with a good menu and a good kitchen. Six months later it serves the same food to half the customers. Nobody read the comment cards. The kitchen was never the problem.

Objective 1.6: value pillars, the boardroom's five words

The household budget argument. We should get a dishwasher convinces nobody. We hand-wash 90 minutes a day; a Rs 40,000 machine returns that time inside five months ends the argument. Same purchase, but the second sentence names a pillar (time), a number (90 minutes) and a horizon (five months). Meera's board hears dishwasher arguments all day. Only numbered pillars survive.

The exam's five pillars, applied to the control tower. A solution is *aligned* when its design choices trace to a pillar with a number:

Pillar	Kaveri number	The design choice it justifies
Efficiency	First touch from 4 hours to under 5 minutes	Routing plus chaining tier for the routine 80 percent
Cost	Model spend at most Rs 6 per exception on average	Small models at the door (E3); big ones only where money moves
Productivity	Dispatchers at 3x throughput, editing not drafting	Draft-plus-gate output design, not auto-send
Performance / SLA	99 percent of drafts within 30 seconds	Parallel carrier checks; agent tier reserved for the tangled few
Transformation	Customs-hold pattern reports become a new pre-clearance service line	The feedback log is a product, not exhaust

Trap: the pillar swap. The reversed trade-off's home ground. The stem fixes one pillar, a 2 second SLA, a cost ceiling, and one option optimises a different, worthier-sounding pillar: maximise accuracy with a 3-model voting ensemble. Wrong. The stated pillar is the contract; the exam pays no marks for nobler physics than the customer ordered.

LAW 5 · Value is a pillar with a number attached. Improves customer experience is a wish. Cuts first-touch from 4 hours to 5 minutes at Rs 6 or less per exception is an architecture requirement, and every Domain 1 option can be tested against it. In 1.6 stems, the option that names measure, number and horizon beats the one that names technology.

Atlas A7: the value-pillar matrix

Boardroom Tuesday. Three candidate architectures scored the only way a 1.6 stem accepts, numerically per pillar. Green meets Meera's target; red breaches it.

Pillar and target	A: one big agent, every ticket	B: tiered, route + chains, agent for 20%	C: rules only, no LLM
Efficiency, first touch at most 5 min	about 2 min, meets	about 1 min, meets	4 h, breaches
Cost, at most Rs 6 per exception	about Rs 22, breaches (agent loops on trivia)	about Rs 4.10, meets	about Rs 0, meets
Productivity, 3x throughput	about 2.8x, meets	about 3.1x, meets	1x, breaches
SLA, 99% of drafts in 30 s	about 91%, breaches (loop latency tail)	about 99.3%, meets	no drafts
Transformation, new service lines	logs exist, unmined	customs pre-clearance line from the feedback log	none

Option B wins not because it is cleverest but because it is the only column with no red cell. **Alignment means no pillar breached, not one pillar maximised.** Option A is the magnificent bridge: it beats B on nothing Meera measures and loses on two pillars. That asymmetry, capable but misaligned, is how 1.6 writes its best distractors.

Trap: the unpriced option. When an option's cells cannot be filled with numbers at all (improves experience, future-proof), it is not a candidate. It is a wish. An option you cannot score against the stated targets loses to any option you can, even a humbler one.

Domain 3 connection

The feedback stage is what the lab's E7 cameras instrument: structured log lines are the raw material of next week's eval set. The SLA pillar's number is the p95 the digest printed. The cost pillar is the 47 paise on the One Ticket bill. Domain 1 names the pillars; Domain 3 is where their numbers come from.

***The one thing to remember.** Four stages, and the missing one is usually feedback or the human gate. Five pillars, each with a number, and the winning option is the one that breaches none, not the one that maximises one.*